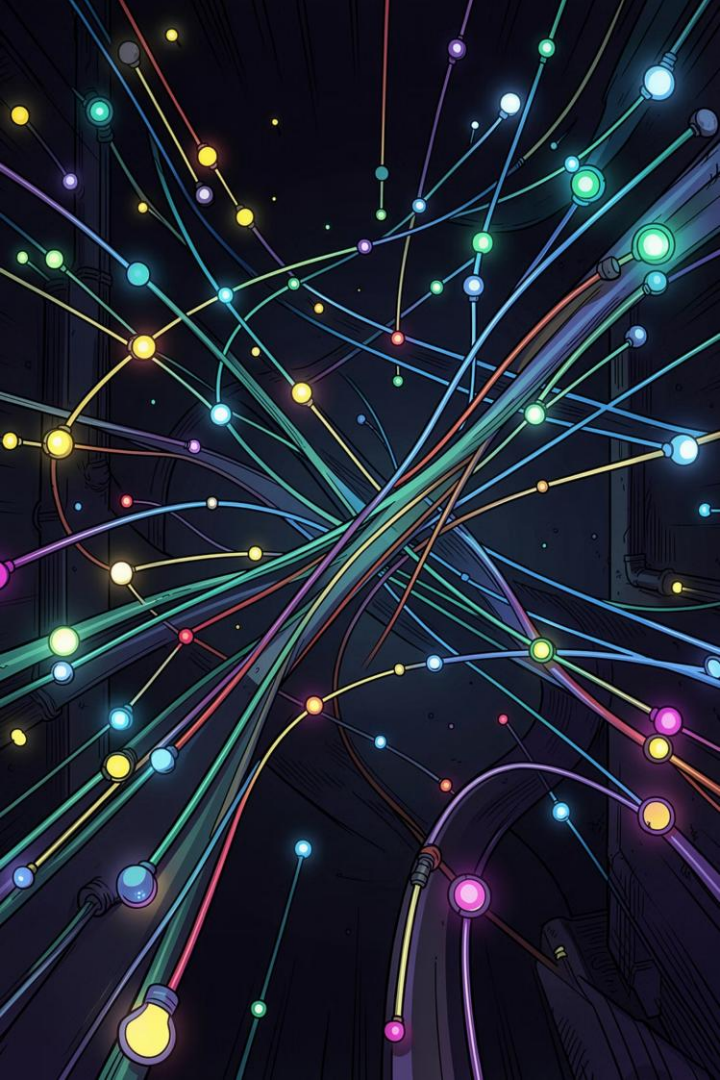


Customer Behaviour Analysis

Using Python, PostgreSQL & Power BI

PROJECT REPORT





Abstract

Extract & Transform

Python libraries: Pandas & SQLAlchemy

Load

Structured ETL pipeline into PostgreSQL

Visualize

Power BI dashboards for customer insights

Introduction



Why It Matters

Customer behaviour analysis helps businesses understand **buying patterns** and **preferences**.

The Approach

This project integrates **Python**, **PostgreSQL**, and **Power BI** into a complete analytics workflow.

Objectives

1

Analyze Purchasing Behaviour

2

Build ETL Pipeline

Python + PostgreSQL

3

SQL-Based Analysis

4

Power BI Dashboards

Interactive reporting



Tools Used



Python

Core language for data processing



Pandas

Data manipulation & transformation



PostgreSQL

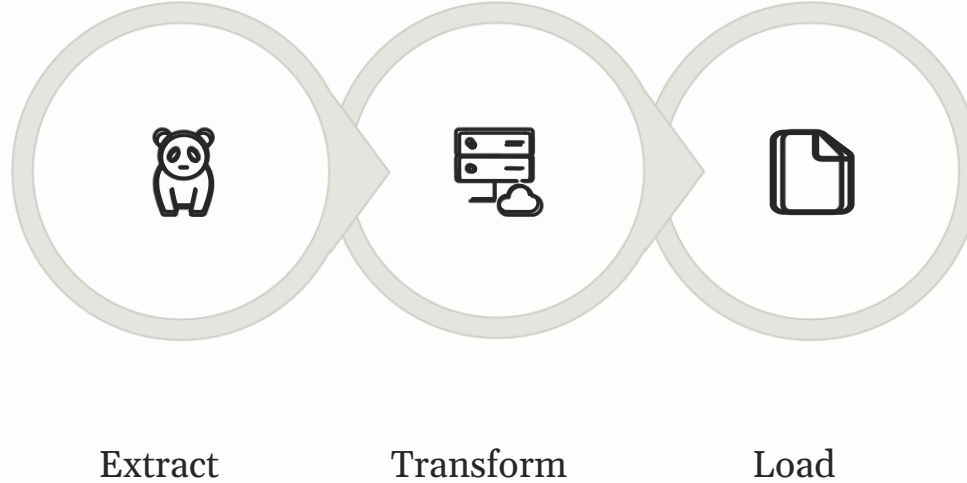
Relational database for storage



Power BI

Interactive dashboards & reporting

ETL Pipeline



Python handles extraction and transformation. PostgreSQL manages loading and storage. Power BI powers reporting and dashboards.

Business Questions Analyzed

→ Revenue: Male vs Female

→ Top-Rated Products

→ Revenue by Age Group

→ Category-Wise Sales

→ Subscription Behaviour

Key Findings

\$59

Avg Purchase

Average purchase amount per transaction

#1

Clothing

Highest-selling category

Top

Young Adults

Highest revenue-contributing age group

Most

Non-Subscribers

Majority of customers were non-subscribers

Power BI Dashboard



Dashboard Components

KPI Cards

Revenue Charts

Category Analysis

Customer Segmentation

Conclusion

This project successfully demonstrated a complete **customer analytics workflow** — from raw data to actionable insights.

Python

Extraction & transformation

PostgreSQL

Structured data storage

Power BI

Interactive dashboards

